Non-Injectivity as a Structural Necessity of Genuine Emergence: A No-Go Theorem for Faithful Emergent Descriptions

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Abstract

We prove that any genuinely emergent (i.e. non-isomorphic effective) physical description is necessarily non-injective and intrinsically compressive. The argument is purely structural: given a surjective projection $\Pi : \Omega \to \mathcal{O}$ such that $\mathcal{O}$ carries no independent microstate-resolving structure beyond what Π provides, and such that the effective level is not structurally isomorphic to the fundamental level, then Π cannot be injective and the associated projection entropy $S_\Pi > 0$. The proof uses only surjectivity, informational completeness of Π for observables, and the non-isomorphism condition; it is independent of any specific physical framework. The result constitutes a *no-go theorem for faithful emergence*: a surjective projection that is both informationally faithful and genuinely emergent cannot exist. As a corollary, descriptive redundancy, non-injectivity of Π, and $S_\Pi > 0$ are co-extensive in any projective emergent framework; any non-trivial gauge structure requires these properties; and any theory admitting a regime where its own description ceases to be valid — such as standard cosmology at $t \lesssim t_{\mathrm{Planck}}$ — is structurally consistent with the presence of a non-injective projection. We provide a categorical reformulation (genuine emergence corresponds to a non-faithful functor) and an informational one (the Kullback–Leibler divergence of any state distribution with respect to its projection lift is strictly positive). The result is illustrated by a concrete spin model and discussed in relation to the projective incompleteness of ΛCDM and the Standard Model, whose fundamental constants are identified as invariants of an unspecified projection Π not accessible from the effective level alone.

Keywords. emergence, non-injectivity, projection, gauge symmetry, information loss, entropy, effective theories.

1 Introduction

Many physical frameworks assume the existence of two distinct levels of description: a fundamental level Ω of underlying configurations, and an effective level $\mathcal{O}$ of observable states. The relationship between these levels is typically encoded in a projection map $\Pi : \Omega \to \mathcal{O}$.

A recurring feature of such frameworks is that Π is many-to-one: distinct configurations $\omega \in \Omega$ may correspond to the same observable state $o \in \mathcal{O}$. This feature is closely related to coarse-graining and renormalisation procedures, where effective descriptions arise through systematic loss of microscopic information [1, 2, 3]. More generally, the relation between emergence and information compression has been discussed in both physical and philosophical contexts [4, 5]. This non-injectivity has been identified as the structural origin of gauge symmetry [17], Bell-inequality violations [18], and black hole entropy [19], and is broadly related to redundancy structures appearing in constrained and coarse-grained descriptions. In each case, however, non-injectivity has been introduced as a feature of a specific model rather than derived as a universal necessity. What is missing across these frameworks is a structural necessity result: while non-injective mappings appear operationally, no general theorem establishes that they are unavoidable for emergence itself.

The present note supplies this missing result. We show that non-injectivity is not a modelling choice but a structural consequence of genuine emergence: any surjective map Π whose observables are entirely defined by Π, and whose effective level is not isomorphic to the fundamental level, must be non-injective. The proof is elementary and framework-independent.

A preliminary observation: standard cosmology as an effective description

Before stating our results, we draw attention to a structural feature of Big Bang cosmology that motivates the theorem.

In the standard cosmological model ΛCDM, an effective spacetime geometry $g_{\mu\nu}$ is well defined and descriptively valid for $t > t_{\text{Planck}}$. For $t \lesssim t_{\text{Planck}}$, the geometric description ceases to be valid: the metric diverges, the Einstein equations break down, and $g_{\mu\nu}$ is not a well-defined object. Yet the large-scale geometry that exists for $t \gg t_{\text{Planck}}$ is stable, continuous, and physically meaningful.

These three facts suggest that $g_{\mu\nu}$ should be regarded as an effective object. A description that fails in a given regime is naturally interpreted as arising from a more fundamental structure whose validity extends beyond that regime. From this perspective, the breakdown of $g_{\mu\nu}$ indicates the limit of applicability of the geometric description, rather than a failure of physical existence.

If this line of interpretation is adopted, one may view $g_{\mu\nu}$ as arising from a more primitive structure Ω, through a projection Π that becomes well-defined when appropriate conditions for projectability are met.

Remark 1 (Heuristic status of this argument)**.** *The implication "$g_{\mu\nu}$ breaks down $\Rightarrow \Pi$ exists and is non-injective" is a heuristic rather than a logical consequence of Theorem 3. Theorem 3 applies to frameworks that explicitly posit Ω and Π; it does not construct them from the mere existence of a singularity. The argument here is interpretive: it identifies the breakdown of $g_{\mu\nu}$ as a structural signal that the geometric description is effective, and* *suggests that any complete successor theory would need to supply the missing Ω and Π. This does not constitute a critique of the standard model, but a reformulation of its domain of validity in structural terms. Once such a successor is posited — and once it satisfies Definitions 1 and 2 — Theorem 3 applies and forces non-injectivity.*

In this sense, any theory that admits a regime where its own description ceases to be valid can be interpreted as an effective description with a restricted domain of applicability. Such a theory does not necessarily specify the underlying level explicitly, but its structure is compatible with an emergent interpretation.

> *Any cosmological framework admitting a regime where spacetime geometry ceases to be well-defined can be consistently interpreted as an effective description, suggesting the existence of a more fundamental level from which geometry arises. The absence of an explicit formulation of this underlying level does not eliminate this interpretation, but leaves it structurally incomplete.*

From this viewpoint, the absence of a geometric description for $t \lesssim t_{\text{Planck}}$ should be understood as delimiting the regime of validity of the spacetime description, rather than as implying the absence of an underlying structure.

If a projection Π relating such a fundamental level Ω to the effective observables $\mathcal{O}$ is introduced and satisfies the conditions of Section 2, Theorem 3 implies that it must be non-injective. The non-injectivity is therefore not a model-dependent feature, but a structural necessity of any complete projective-emergent framework.

Key message. A fully injective emergent description is a contradiction in terms under Definitions 1 and 2: if Π were injective, the effective level would be isomorphic to the fundamental level, and there would be no emergence — only a relabelling.

2 Framework and Definitions

Let Ω be a space of fundamental configurations and $\mathcal{O}$ a space of observable states. We consider a surjective map $\Pi : \Omega \to \mathcal{O}$.

Definition 1 (Observables defined by Π)**.** *The observables of $\mathcal{O}$ are* defined by Π *if $\mathcal{O}$ carries no independent microstate-resolving structure: every distinction present in $\mathcal{O}$ is a distinction already present in Ω and transported by Π, with no additional resolving power assigned to $\mathcal{O}$ by an external postulate. Equivalently, two configurations $\omega, \omega' \in \Omega$ are observationally indiscernible if and only if* $\Pi(\omega) = \Pi(\omega')$.

This condition is the minimal formalisation of what it means for observables to *arise from* a more fundamental level rather than being postulated independently. It is weaker than requiring that every structure on $\mathcal{O}$ be explicitly constructed from Ω: it only requires that $\mathcal{O}$ introduces no new resolving power beyond what Π already provides. Equivalently, Definition 1 can be rephrased as: Π *is informationally complete for observables* — meaning that every distinction $\mathcal{O}$ can draw is already drawn by Π, and $\mathcal{O}$ adds no further discriminating capacity of its own. Any theory that assigns independent microstate-resolving attributes to $\mathcal{O}$ is, by that act, a two-level theory with autonomous observable structure, not an emergent one. Frameworks where $\mathcal{O}$ carries independent structure — such as standard quantum mechanics or general relativity, which directly postulate a Hilbert space or a spacetime manifold — lie outside this class by construction; see Section 6 for a precise discussion.

Example 1 (Spin model)**.** *Let* $\Omega = \{-1, +1\}^N$ *be the configuration space of* N *Ising spins, and let* $\mathcal{O} = \mathbb{R}$ *with* $\Pi(\sigma) = N^{-1} \sum_i \sigma_i$ *the mean magnetisation. Distinct spin configurations with the same mean magnetisation are observationally indiscernible:* $\mathcal{O}$ *carries no structure that distinguishes them. Definition 1 is satisfied. Condition (E3) holds since* $|\mathcal{O}| \ll |\Omega|$ *for large* N*. Theorem 3 then guarantees* $S_\Pi > 0$*, which is immediate: for any magnetisation value* $m \in (-1, 1)$*, the fibre* $\Pi^{-1}(m)$ *has exponentially many elements. The non-injectivity is not merely observed here; it is a structural necessity of the emergent description.*

Definition 2 (Effectively non-trivial description)**.** *The description is* effectively non-trivial *if:*

(E1) ***Distinguishability:*** Π *is not constant — there exist* $\omega_1, \omega_2 \in \Omega$ *such that* $\Pi(\omega_1) \neq \Pi(\omega_2)$*.*

(E3) ***Structural non-isomorphism:*** $\mathcal{O} \not\cong \Omega$ *as structured spaces, where "structured space" refers to the full set of physically relevant structures induced by* Π *(topology, algebraic relations, observables, or dynamical laws).*

Remark 2. *Condition (E3) is the formal statement that the effective level is genuinely distinct from the fundamental level. An isomorphism* $\mathcal{O} \cong \Omega$ *would mean that* $\mathcal{O}$ *is merely a relabelling of* Ω*, not an emergent structure. Note that (E2) — the statement that* Π *is non-injective — is deliberately absent from Definition 2: including it would make the main theorem circular.*

We also recall the standard projection entropy associated with Π. Given a non-degenerate measure μ on Ω, define $\nu(o) = \mu(\Pi^{-1}(o))$ for each $o \in \mathcal{O}$. The *projection entropy* is

$$S_\Pi \equiv -\sum_{o \in \mathcal{O}} \nu(o) \log \nu(o). \tag{1}$$

$S_\Pi = 0$ if and only if Π is (almost everywhere) injective; $S_\Pi > 0$ if and only if at least one fibre $\Pi^{-1}(o)$ has measure greater than that of a single point [18, 17].

3 Main Result

The proof of Theorem 3 rests entirely on the following lemma, which we state and prove separately to make the logical hinge explicit.

Lemma 2 (Injective projection is an isomorphism)**.** *Let* $\Pi : \Omega \to \mathcal{O}$ *be surjective and injective, with observables defined by* Π *in the sense of Definition 1. Then* Π *is a structural isomorphism* $\mathcal{O} \cong \Omega$*.*

Proof. Since Π is surjective and injective, it is bijective. Since all structure on $\mathcal{O}$ is induced from Ω by Π (Definition 1), a bijection that transports all structure is a structure-preserving isomorphism. Hence $\mathcal{O} \cong \Omega$. □

Remark 3 (Why this lemma is the pivot)**.** *Lemma 2 is the single logical hinge of the entire paper. Its content is deceptively simple: if nothing is lost under* Π *(injectivity) and everything in* $\mathcal{O}$ *comes from* Ω *(Definition 1), then* $\mathcal{O}$ *and* Ω *are the same structured space under different names. There is no emergence, only relabelling. Condition (E3) of Definition 2 forbids precisely this — and thereby forces non-injectivity.*

Theorem 3 (Compression necessity)**.** *Let $\Pi : \Omega \to \mathcal{O}$ be surjective, with observables defined by Π in the sense of Definition 1 and description effectively non-trivial in the sense of Definition 2. Then Π is necessarily non-injective, and for any non-degenerate measure μ on Ω,*

$$S_\Pi > 0.$$

Proof. Suppose for contradiction that Π is injective. By Lemma 2, $\mathcal{O} \cong \Omega$ as structured spaces. This contradicts condition (E3). Therefore Π is not injective.

Since Π is not injective, there exists $o \in \mathcal{O}$ such that $|\Pi^{-1}(o)| > 1$. For any non-degenerate measure μ, the weight $\nu(o) \in (0, 1)$ for this o, and $-\nu(o) \log \nu(o) > 0$. Hence $S_\Pi > 0$. □

4 Gauge Structure and Co-extensivity

Proposition 4 (Gauge requires non-injectivity)**.** *In any framework where observables are defined by Π (Definition 1), any non-trivial gauge structure in $\mathcal{O}$ requires Π to be non-injective, and hence $S_\Pi > 0$.*

Proof. If Π were injective, the holonomy group of Π would be trivial ($G_\Pi = \{e\}$), every connection would be flat, and no non-trivial gauge field could emerge from the bundle structure of Π alone. Any non-trivial gauge field would then require an independent postulate at the level of $\mathcal{O}$, violating Definition 1. □

Remark 4 (On the converse)**.** *The converse of Proposition 4 does not hold in general: non-injectivity does not automatically produce gauge structure. For the redundancy encoded in non-trivial fibres to organise into a gauge symmetry, additional geometric hypotheses are required: the fibres $\Pi^{-1}(o)$ must form a principal bundle over $\mathcal{O}$ with a non-trivial structure group G_Π admitting a connection. This is a result supplementary to Theorem 3, not a consequence of it.*

Corollary 5 (Co-extensivity)**.** *In any projective emergent framework satisfying the hypotheses of Theorem 3,*

$$\textit{descriptive redundancy} \iff \Pi \textit{ non-injective} \iff S_\Pi > 0,$$

and any non-trivial gauge structure requires these properties:

$$\textit{non-trivial gauge} \implies \Pi \textit{ non-injective} \iff S_\Pi > 0.$$

Under additional geometric hypotheses on the fibre organisation of Π (principal bundle structure, admissible connection, non-trivial holonomy), the redundancy may in turn manifest as a gauge structure.

This perspective is consistent with the standard interpretation of gauge symmetry as a redundancy of description, as formalised in constrained Hamiltonian systems [6] and relational approaches to observables [7].

5 Three-Level Structure

The results of Sections 3 and 4 establish a hierarchy with three distinct levels of increasing specificity.

Universal level. Theorem 3 holds for any surjective Π satisfying Definitions 1 and 2. No dynamical, geometric, or physical hypothesis is required beyond the structural conditions on Π. The conclusion — Π non-injective, $S_\Pi > 0$ — is a structural constraint on any framework that takes emergence seriously.

Intermediate level. If the non-trivial fibres of Π are organised geometrically — forming a principal G_Π-bundle with admissible connection — then the projective redundancy can support a gauge structure with holonomy group G_Π. This is an additional geometric hypothesis, not a consequence of Theorem 3 alone.

Cosmochrony level. In the Cosmochrony relational framework [17], the intermediate hypothesis is explicitly realised: the projection $\Pi : \mathcal{C}_\chi \to \mathcal{M}$ defines a principal bundle with fibre S^3, yielding $G_\Pi \supseteq \mathrm{SU}(2) \times \mathrm{U}(1)$. The resulting gauge structure underlies the emergence of electroweak interactions, while the Born–Infeld saturation bound selects the unique admissible infrared dynamics [19]. Bell-type correlations arise as the observable signature of $S_\Pi > 0$ [18]. The route from non-injectivity to the specific group SU(2), via Born–Infeld parity, Weil representations, and the Hurwitz classification of real normed division algebras, is detailed in Appendix A.

6 Scope and Limits of the Result

Theorem 3 is a conditional result. Its conclusion holds whenever the hypotheses of Definitions 1 and 2 are satisfied. It is therefore essential to identify the class of theories to which it applies and the class to which it does not.

Theories outside the scope. Standard physical frameworks — quantum mechanics, general relativity, standard-model field theories — are not covered by Theorem 3. These theories posit their fundamental objects (Hilbert space, spacetime manifold, gauge fields) directly at the level $\mathcal{O}$, without deriving them from a more primitive space Ω via a projection. The question of non-injectivity does not arise in such frameworks: there is no projection Π to be injective or otherwise. The theorem is silent about them.

Loop quantum gravity (LQG) deserves explicit discussion, as it occupies an intermediate position [8, 9]. LQG describes the quantum geometry of the universe — spin networks and spin foams — as fundamental objects that are themselves part of the observable universe, not external to it. It does not posit a pre-geometric substrate Ω from which observables are projected; it claims to *be* the fundamental level. In this sense LQG is not emergent: it does not fit the projective structure of Definition 1, and Theorem 3 does not apply to it as a fundamental theory.

The relevant demarcation criterion is therefore not the presence of a singularity but an ontological one:

> *Theorem 3 applies naturally to frameworks that posit a level Ω structurally external to the observable universe, from which observables are projected via* Π. *A theory that describes the universe at a finer scale — however fine — without positing such an external level falls outside this class.*

LQG describes the universe at the Planck scale; it does not describe what lies outside or before the universe. Cosmochrony, by contrast, posits a pre-geometric substrate $\mathcal{C}_\chi$ that is pre-spatial, pre-temporal, and structurally prior to the observable universe: it is explicitly in the class to which the theorem applies.

Note, however, that the semi-classical limit of LQG — the projection $\Pi_{\text{sc}} : \mathcal{H}_{\text{LQG}} \to g_{\mu\nu}$ that recovers classical general relativity — is itself a non-injective map within the class of Definition 1, with $\Omega = \mathcal{H}_{\text{LQG}}$ and $\mathcal{O} = \{g_{\mu\nu}\}$. Theorem 3 applies to this projection: general relativity is genuinely emergent from LQG, and its projection is necessarily non-injective. The two claims — LQG is not emergent as a fundamental theory, GR is emergent from LQG — are consistent.

Remark 5 (Multi-valued maps lie outside the scope)**.** *Theorem 3 assumes a single-valued surjection $\Pi : \Omega \to \mathcal{O}$. Multi-valued maps $\Phi : \Omega \rightrightarrows \mathcal{O}$, in which a single configuration $\omega \in \Omega$ may be associated with several distinct observables, fall outside the framework.*

The exclusion is independent of injectivity and follows directly from Definition 1. If $\Phi(\omega)$ contains distinct elements $o_1 \neq o_2$, then $\mathcal{O}$ resolves a distinction — between o_1 and o_2 — that is not resolved by the fundamental description at ω. The observable level then carries a strictly finer discriminating structure than the fundamental level, which is incompatible with the requirement that $\mathcal{O}$ carry no independent microstate-resolving structure.

Such generative maps describe a structurally different relationship between levels: one in which the effective description is more expressive than the fundamental one. Whether they admit structural necessity results analogous to Theorem 3 is an open question.

Theories within the scope. Theorem 3 applies to any framework where the effective description is genuinely derived from a finer-grained level via projection. This includes, in particular: renormalisation group (RG) approaches, where the projection $\Pi_\Lambda : \Omega_{\text{UV}} \to \mathcal{O}_{\text{IR}}$ integrates out modes above the scale Λ and is non-injective by construction; causal set models, where macroscopic geometry is an invariant of sets of micro-configurations and the mapping is many-to-one; spin-foam and spin-network models, where the effective geometrical observables arise from averaging over discrete micro-structures; and relational frameworks such as Cosmochrony [17], where the full observable structure is explicitly derived from a pre-geometric substrate via a projection Π.

In each of these cases, the theorem does not prove non-injectivity from nothing: it shows that the non-injectivity already implicit in the framework's own definition of "effective description" is a logical necessity, not a contingent modelling choice.

The key distinction. The result isolates a precise logical boundary:

> *Within the present projective-emergent setting, injectivity of* Π *is equivalent to the absence of genuine emergence.*

If Π is injective, the effective level is isomorphic to the fundamental level: nothing is lost, nothing is emergent. If the effective level is genuinely distinct, something is necessarily lost, and that loss is precisely $S_\Pi > 0$.

7 Categorical and Informational Reformulations

Two reformulations of Theorem 3 strengthen its connections to existing mathematical frameworks.

Categorical form. Let **Fund** be the category whose objects are spaces of fundamental configurations and whose morphisms are dynamically compatible maps, and let **Obs** be the category of observable spaces. The projection Π defines a functor $F_\Pi : \mathbf{Fund} \to \mathbf{Obs}$.

Recall that a functor is *faithful* if it distinguishes all morphisms that the source category distinguishes; faithfulness is the categorical analogue of injectivity.

Theorem 6 (Categorical form of compression necessity)**.** *Under the hypotheses of Theorem 3, the functor* $F_\Pi : \mathbf{Fund} \to \mathbf{Obs}$ *is necessarily non-faithful. Conversely, a faithful functor from* **Fund** *to a non-equivalent* **Obs** *cannot exist.*

Proof. If F_Π were faithful, the induced map on morphisms would be injective at every hom-set, which at the level of objects recovers the injectivity of Π. By Lemma 2, this forces $\mathcal{O} \cong \Omega$, i.e. $\mathbf{Obs} \simeq \mathbf{Fund}$, contradicting condition (E3). Hence F_Π cannot be faithful. The converse follows immediately: a faithful functor to a non-equivalent category is a contradiction in the same sense. □

Remark 6. *Theorem 6 identifies genuine emergence with the failure of faithfulness in the categorical sense. Faithful functors preserve all distinctions of the source; non-faithful functors compress them. Emergence is precisely this compression, now stated as a categorical necessity.*

Informational version. Let p be a probability distribution on Ω and $q = \Pi_* p$ its pushforward on $\mathcal{O}$. The Kullback–Leibler divergence of p with respect to the lifted distribution $\Pi^* q$ measures the information lost under Π:

$$D_{\mathrm{KL}}(p \,\|\, \Pi^* q) \;=\; \sum_{\omega\in\Omega} p(\omega) \log \frac{p(\omega)}{q(\Pi(\omega))}. \tag{2}$$

If Π is injective, $p(\omega) = q(\Pi(\omega))$ for all ω and $D_{\mathrm{KL}} = 0$. Theorem 3 implies:

Corollary 7 (Informational lower bound)**.** *In any effectively non-trivial emergent framework, there exists a distribution* p *on* Ω *such that* $D_{\mathrm{KL}}(p \,\|\, \Pi^* q) > 0$*. The projection entropy* S_Π *provides a distribution-independent lower bound on the structural information loss.*

The projection entropy S_Π defined in Eq. (1) is the distribution-independent part of this loss: it captures the structural compression built into Π independently of any particular state. D_{KL} then adds the state-dependent contribution. Together, they quantify two distinct aspects of the same fundamental compression.

The informational reading also connects to the renormalisation group. The Wilsonian RG projection Π_Λ is non-injective by construction [1]: the same low-energy observable corresponds to infinitely many UV completions. Corollary 7 says that any such effective field theory has $S_{\Pi_\Lambda} > 0$ and $D_{\mathrm{KL}} > 0$: the loss of UV information is not a limitation to be overcome but a structural property of the emergent description itself.

This interpretation aligns with the information-theoretic view of statistical mechanics and inference, where macroscopic descriptions necessarily involve a loss of microscopic detail [2].

8 Discussion

The central result admits a concise reading:

> *Any genuinely emergent physical description necessarily involves information compression. A fully injective projection would reduce emergence to a mere reparametrisation.*

Equivalently, Theorem 3 can be read as a *no-go theorem for faithful emergence*: a surjective projection that is both informationally faithful (injective) and genuinely emergent ($\mathcal{O} \ncong \Omega$) cannot exist. One of the two conditions must be abandoned; genuine emergence requires abandoning injectivity.

While the connection between emergence and information loss is widely recognised in the context of coarse-graining and effective field theories, the present result sharpens this intuition into a structural necessity: it does not rely on a specific dynamical or statistical framework, but follows from minimal assumptions on the relation between levels of description.

The hypotheses are minimal. We do not assume a metric, a Hilbert space, a Lagrangian, or any physical dynamics. We require only that the effective description be defined by projection (Definition 1) and genuinely distinct from the fundamental level (condition E3).

The connection to information theory is direct. The projection entropy S_Π is not an epistemic quantity: it does not measure ignorance about which $\omega \in \Omega$ is the true configuration. It measures an objective structural property of Π — the degree to which Ω distinctions are compressed out of $\mathcal{O}$. Theorem 3 establishes that this compression is not optional: in any genuine emergent framework, $S_\Pi > 0$ is a structural necessity, not a contingent feature.

The result also clarifies the status of gauge symmetry. Gauge invariance has long been understood as a redundancy of description. Proposition 4 and Remark 4 make this precise in the projective setting: redundancy is equivalent to non-injectivity, which is itself forced by the structure of genuine emergence. Gauge symmetry is therefore not a fundamental feature of nature but a necessary organisational consequence of any emergent description that is rich enough to carry a principal bundle structure.

Projective incompleteness of standard physics. The frameworks ΛCDM and the Standard Model describe the structure of $\mathcal{O}$ with extraordinary precision, but they specify neither the fundamental level Ω nor the projection $\Pi : \Omega \to \mathcal{O}$. Yet three independent mechanisms within these frameworks each admit a natural reading in terms of a non-injective projection.

(i) Cosmological singularity. The metric $g_{\mu\nu}$ ceases to be defined at $t \lesssim t_{\text{Planck}}$, which is structurally consistent with the geometry being an effective description arising from a more primitive level. If such a level Ω and projection Π are posited and satisfy Definitions 1 and 2, Theorem 3 forces Π to be non-injective.

(ii) Renormalisation group. The Standard Model can be naturally interpreted as a Wilsonian effective field theory [1] with UV cutoff Λ. Under this interpretation, the map $\Pi_\Lambda : \Omega_{\text{UV}} \to \mathcal{O}_{\text{IR}}$ is non-injective: infinitely many UV completions yield the same infrared observables. Similar compressive structures appear in classical statistical mechanics through phase-space coarse-graining, where macroscopic thermodynamic variables correspond to vast equivalence classes of microstates.

(iii) Quantum measurement. The transition from quantum states to classical outcomes admits a natural description as a projection $\Pi_{\text{meas}} : \mathcal{H} \to \mathcal{O}_{\text{class}}$ mapping distinct quantum states to the same classical outcome. The measurement problem can then be read, in the present language, as the question of what determines the structure of Π_{meas} and why it is non-injective.

These three readings are independent of any hypothesis specific to Cosmochrony. They illustrate that the projective structure identified in Theorem 3 is not exotic: it appears wherever an effective description is obtained from a finer-grained level. What is missing in these frameworks is a structural necessity result: while non-injective mappings appear operationally in each case, no general theorem establishes that they are unavoidable for emergence itself. Theorem 3 supplies this missing result. The following table identifies what is described and what is absent in standard physics.

Level	Standard physics	Missing
Observable space $\mathcal{O}$	fully described	—
Fundamental space Ω	absent	unknown structure
Projection Π	absent	law of projection
Fibres $\Pi^{-1}(o)$	absent	microscopic multiplicity

The quantities that standard physics cannot derive — the fine-structure constant α, the electron-to-proton mass ratio m_e/m_p, the CKM mixing angles — can be naturally interpreted as invariants of Π that $\mathcal{O}$ alone cannot determine: under this interpretation, they encode properties of the fibres $\Pi^{-1}(o)$ that are invisible to any description confined to the effective level.

The incompleteness is not empirical but structural:

Standard physics omits the map defining its observables.

This imposes a precise constraint on any candidate unification or quantum-gravity programme: a theory that remains within $\mathcal{O}$ — however sophisticated — does not complete the description; it extends it. Any theory that genuinely completes standard physics must specify Ω, Π, and the non-injective structure of Π.

Singularity elimination does not restore completeness. A natural response is to propose a framework that eliminates the initial singularity entirely — loop quantum cosmology (LQC), pre-Big Bang scenarios, bounce cosmologies, or emergent-universe models — on the grounds that if $g_{\mu\nu}$ is everywhere well defined, the cosmological argument of Section 1 does not apply. This response is correct as far as it goes, but it confuses eliminating a *signal* of incompleteness with eliminating the incompleteness itself.

Consider the four main classes. LQC replaces the singularity by a quantum bounce at Planck density [10, 11]. It has a partial Ω — spin-network states — and a many-to-one projection to the effective metric. It is more advanced than ΛCDM in specifying Ω, but Π remains unspecified: the effective dynamics is postulated, not derived from the projection structure. Pre-Big Bang scenarios (string cosmology [12], ekpyrotic [13], conformal cyclic [14]) displace the singularity into a preceding phase, but that phase is itself described by an effective geometry — the question of its own Ω and Π is deferred, not answered. Bounce cosmologies with everywhere-regular $g_{\mu\nu}$ avoid the cosmological argument directly; they remain incomplete through the renormalisation and measurement non-injectivities of mechanisms (ii) and (iii) above. Emergent-universe models (Hartle–Hawking [15], tunnelling from nothing [16]) replace the singularity by a quantum-to-classical transition — which is itself a non-injective projection from $\mathcal{H}$ to a classical geometry.

In every case, the structure of the argument is the same:

Eliminating the singularity removes one signal of projective incompleteness. It does not supply Ω, Π, or the non-injective structure of Π*. A theory without a singularity but without a fundamental level is simply a theory whose limit of validity has not yet been reached within the observed range.*

Completeness in the projective sense requires specifying what lies above $\mathcal{O}$, not merely extending $\mathcal{O}$ to a regime where it no longer breaks down.

The three reformulations — structural (Theorem 3), categorical (Theorem 6), and informational (Corollary 7) — are equivalent in content but address different audiences and connect to different existing bodies of work. Their convergence on the same conclusion strengthens the claim that non-injectivity is a universal structural feature of emergent descriptions, not an artefact of any particular formalism.

On the non-triviality of the result. A natural objection is that non-injectivity is "obvious" in any coarse-graining procedure. This objection conflates two distinct claims. It is indeed obvious that a specific coarse-graining — such as the spin model of Example 1 — produces a non-injective map. What Theorem 3 establishes is different: it shows that *any* framework satisfying Definitions 1 and 2 must be non-injective, independently of the specific form of Π. The theorem is a constraint on the class of theories, not an observation about a particular model. Its content is the logical equivalence genuine emergence $\Leftrightarrow$ non-injectivity within the projective-emergent setting, which rules out the possibility of a faithful emergent description — something that is not guaranteed by pointing to examples of coarse-graining. The present result does not introduce a new model of emergence, but establishes a structural constraint that any such model must satisfy: within the present projective-emergent framework, emergence without non-injectivity is impossible.

The core equivalence, within the present projective-emergent setting, can be stated in a single line:

$$\text{genuine emergence} \iff \text{information loss} \iff \Pi \text{ non-injective} \iff S_\Pi > 0. \quad (3)$$

The left-to-right direction is Theorem 3; the right-to-left direction follows from the fact that any non-injective Π whose fibres are non-trivial defines a genuinely distinct effective level by the identification of distinct configurations. A fully injective effective theory is not an emergent theory but a renaming of the fundamental level.

A On the Route from Non-Injectivity to $\mathrm{SU}(2)$

Theorem 3 guarantees the existence of non-trivial fibres but says nothing about their form. The identification of the structure group G_Π with $\mathrm{SU}(2)$ rather than $\mathrm{SO}(3)$ or another compact group is a separate result, established in the Cosmochrony programme via a chain of constraints absent from the universal theorem.

The route proceeds in four steps. First, the Born–Infeld parity constraint [21] fixes the minimal fibre structure to the involution $c \leftrightarrow q - c$ on the Weil representation space of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. Second, the requirement that the representation algebra be associative, combined with the Hurwitz classification of real normed division algebras ($\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, $\mathbb{O}$), selects $\mathbb{H}$ as the minimal associative non-commutative structure [22]. Third, the admissible neutral sector has exactly $\dim_{\mathbb{R}} \mathrm{Im}\,\mathbb{H} = 3$ independent directions — the content of $\Sigma_c(n_3) = 3$ in [22] — and the groups realising this sector form the set $\{Q_8, \mathrm{Dic}_n, 2O, 2I\}$, all subgroups of $\mathrm{SU}(2)$ and not of $\mathrm{SO}(3)$ [17]. The exclusion of $\mathrm{SO}(3)$ is structural: quotienting the central $\mathbb{Z}_2$ of $\mathrm{SU}(2)$ destroys the 4π-periodicity required for spinorial holonomy. Fourth, in the continuum limit, the chain $Q_8 \hookrightarrow 2I \hookrightarrow \mathrm{LPS}_p \hookrightarrow \mathrm{SU}(2)$ converges to $\mathrm{SU}(2)$ as the unique spectral attractor [17].

These steps use Born–Infeld dynamics, Weil representations on Heisenberg groups, and the Hurwitz theorem — none of which appears in Theorem 3. The universal theorem supplies the necessary condition (non-trivial fibres); the Cosmochrony constraints supply the sufficient conditions that fix their geometry to $S^3 \cong \mathrm{SU}(2)$. This selection is model-dependent: other emergent frameworks may yield different fibre geometries and different structure groups. The two results are complementary and non-redundant.

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